

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech.

Branch: Electronics &telecommunication Engineering

Semester: V

Subject Code & Name: Electromagnetic Field Theory BTEXC501_Y18

Max Marks: 60

Date: 06/02/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The value for permittivity of free space is ϵ_0	Understand,	1
	a. 8.854×10^{-12} F/m b. 8.254×10^{-12} F/m c. 8.854×10^{-10} F/m d. 8.854×10^{12} F/m	Remember	
2	Coulomb's Law describes the force between two-point charges and is proportional to	Understand	1
	a. The product of the charges b. The square of the charges c. The inverse square of the distance between the charges d. The distance between the charges		
3	The Biot-Savart Law is used to calculate	Understand	1
	a. The electric field due to a moving charge b. The magnetic field due to a current c. The potential due to a static charge d. The energy stored in a magnetic field		
4	According to Ohm's Law, the current density J is related to the electric field E by	Understand	1
	a. $J = \sigma E$ b. $J = \mu E$ c. $J = \epsilon E$ d. $J = \rho E$		
5	Example of spherical system in the following is	Understand	1

	a. charge in space	b. charge in dielectric	c. charge in box	d. charge in system		
6	In a time-harmonic electric field, the displacement current density J is related to:				Understand	1
	a. The rate of change of the electric field	b. The conductivity of the medium	c. The magnetic field intensity	d. The electric potential		
7	The characteristic impedance of a transmission line is denoted by				Understand	1
	a. Z_0	b. $Z_L - Z_0$	c. $Z_0 - Z_L$	d. Z_L		
8	For maximum power transfer, the load impedance Z_L must be				Understand	1
	a. Equal to the characteristic impedance Z_0	b. Greater than the characteristic impedance Z_0	c. Less than the characteristic impedance Z_0	d. Zero		
9	The Smith Chart is used for:				Understand	1
	a. Measuring power in the transmission line	b. Calculating the frequency response	c. Impedance matching and reflection coefficient analysis	d. Determining the electric field in the line		
10	The Poynting theorem describes the:				Understand	1
	a. Flow of electromagnetic energy in a wave	b. Reflection of electromagnetic waves at boundaries	c. Polarization of electromagnetic waves	d. Power dissipation in a resistive medium		
11	In a uniform plane wave, the electric field and magnetic field are:				Understand	1
	a. Perpendicular to each other and to the direction of wave propagation	b. Parallel to each other and to the direction of wave propagation	c. Parallel to the direction of wave propagation	d. Perpendicular to the direction of wave propagation only		
12	The wave equation for an electromagnetic wave in free space is derived from				Understand	1

	a. Maxwell's equations	b. The principle of superposition	c. Coulomb's Law	d. The Helmholtz equation		
Q. 2	Solve the following.					12
A)	State Colomb's Law. By Using Coulomb's Law, calculate the electrostatic force between two-point charges, $Q_1=5\mu\text{C}$ and $Q_2= 3\mu\text{C}$ separated by a distance of 0.2 m.				Understand, Remember	6
B)	Explain Point form of Gauss's Law.				Understand	6
Q.3	Solve the following.					12
A)	State Biot-Savart Law and derive the expression for the magnetic field intensity due to infinite long filament.				Understand, Remember	6
B)	Explain the Ampere's Circular Law.				Understand	6
Q. 4	Solve Any Two of the following.					12
A)	State and explain the Maxwell's equations.				Understand	6
B)	Derive the expression for the electric and magnetic fields of a plane wave in free space.				Understand, Remember	6
C)	State Faraday's Law of Induction.				Understand, Remember	6
Q.5	Solve Any Two of the following.					12
A)	Define: 1) Propagation constant 2) VSWR				Understand	6
B)	Write short note on smith chart.				Understand	6
C)	Explain Impedance Matching in transmission lines using a neat diagram.				Understand, apply	6
Q. 6	Solve Any Two of the following.					12
A)	State and derive pointing theorem.				Understand, Remember	6

B)	Define polarization and differentiate linear, circular, and elliptical polarization (any 4 points)	Understand	6
C)	With the help of diagram explain the reflection of plane wave.	Understand, apply	6
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Q. 1 Objective type questions. (Compulsory Question)		12
1 The surface integral of a vector field represents:		1
a) circulation b) divergence c) flux d) curl		
2 According to Coulomb's law, the force between two charges is:		1
a) independent of the distance b) directly proportional to the distance		
c) inversely proportional to the square of the distance d) directly proportional to the square of the distance		
3 Gauss's law is useful in calculating electric fields for _____ type of charge distributions		1
a) random b) symmetrical c) asymmetrical d) time-dependent		
4 The SI unit of magnetic flux density is:		1
a) Tesla b) Weber c) Ampere/meter d) Henry		
5 The tangential component of B across a boundary:		1
a) Is discontinuous b) Is zero		
c) Is continuous d) Depends on the current density		
6 Which Maxwell's equation relates to Faraday's law of induction?		1
a) $\nabla \cdot \mathbf{E} = \rho / \epsilon_0$ b) $\nabla \cdot \mathbf{B} = 0$ c) $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ d) $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 (\partial \mathbf{E} / \partial t)$		
7 The displacement current term in Maxwell's equations is introduced to:		1
a) Account for magnetic monopoles b) Eliminate the electric field		
c) Replace conduction current d) Satisfy the continuity equation for time-varying fields		
8 On a Smith Chart, a clockwise movement represents:		1
a) Moving toward the load b) Moving away from the load		
c) No change in the distance from the load d) Reflection coefficient approaching zero		
9 In phasor form, Maxwell's curl equation $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ becomes:		1
a) $\nabla \times \mathbf{E} = -j\omega \mathbf{B}$ b) $\nabla \times \mathbf{E} = j\omega \mathbf{B}$ c) $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ d) $\nabla \times \mathbf{E} = 0$		
10 A uniform plane wave in a lossless dielectric medium satisfies the relation:		1
a) $\nabla \cdot \mathbf{E} = 0$ b) $\nabla \cdot \mathbf{B} = 0$ c) $\nabla \cdot \mathbf{E} = -\nabla \cdot \mathbf{B}$ d) Both a) and b)		
11 For oblique incidence, the Brewster angle is the angle at which:		1
a) Reflection coefficient is maximum b) Transmission coefficient is zero		
c) Reflection coefficient is zero for parallel polarization d) Electric field is perpendicular to the boundary		
12 The angle θ between $\mathbf{A} = 3\mathbf{a}_x - 4\mathbf{a}_y = 3$ and $\mathbf{B} = 6\mathbf{a}_x + 8\mathbf{a}_y$ is:		1
a) 0° b) 90° c) 120° d) 180°		

Q. 2 Solve the following.		12
A) Given vectors $\mathbf{P} = 6\mathbf{a}_x + 2\mathbf{a}_y$ and $\mathbf{Q} = \mathbf{a}_x - 2\mathbf{a}_y$, calculate the following:	Apply	6
(a) $\mathbf{P} \times \mathbf{Q}$		
(b) $3\mathbf{P} + 2\mathbf{Q}$		
(c) angle between $\mathbf{P}$ and $\mathbf{Q}$		
B) State and prove stoke's Theorem.	Remember	6
Q.3 Solve the following.		12
A) Derive the continuity equation for charge conservation. Explain the significance of the equation in the context of current flow in conductors.	Apply	6
B) State the Biot-Savart Law, calculate the magnetic field at a point due to a current I flowing through a long straight wire.	Apply	6
Q. 4 Solve Any Two of the following.		12
A) State Faraday's Law of Induction. Derive the mathematical expression for the induced emf in a coil of wire when subjected to a changing magnetic field.	Understand	6
B) State and explain the gradient of a scalar field and find the gradient of the scalar field $\mathbf{U} = x^2 y + x y z$	Understand	6
C) Derive the boundary conditions for the electric and magnetic fields at the interface between two different media.	Analyze	6
Q.5 Solve Any Two of the following.		12
A) Derive the general wave equations for voltage and current on a transmission line. Discuss the role of the characteristic impedance in these equations.	Analyze	6
B) Define (i) VSWR and (ii) reflection coefficient. A transmission line with a characteristic impedance of $Z_0 = 50\Omega$ is terminated with a load impedance of $Z_L = 75\Omega$. 1. Calculate the reflection coefficient at the load. 2. Determine the VSWR for this system.	Apply	6
C) Explain the concept of the Smith Chart and its application in the analysis of transmission lines.	Understand	6
Q. 6 Solve Any Two of the following.		12
A) State Maxwell's equations in phasor form for time-harmonic fields. Discuss the significance of each equation in describing the behavior of electric and magnetic fields in space and time.	Understand	6
B) Explain the concept of polarization in electromagnetic waves. Differentiate between linear, circular, and elliptical polarization.	Understand	6
C) A uniform plane wave of 75 Hz propagating through Copper conductor having $\sigma = 7.2 \times 10^7 \text{ U/m}$, $\epsilon_r = 1$, $\mu_r = 1$. Find (i) attenuation constant, (ii) phase constant, (iii) propagation constant, (iv) intrinsic impedance, (v) wavelength and (vi) phase velocity.	Evaluate	6

*** End ***